

PHYSICS MINI PROJECT

SEMICONDUCTOR PHYSICS

Fundamentals, Energy Bands, Doping, PN Junction and Applications

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1. Introduction

A semiconductor is a material whose electrical conductivity lies between that of a good conductor and an insulator. Its conductivity can be controlled by temperature, light, electric fields and, most importantly for electronic devices, by introducing very small amounts of impurities through a process called doping. This controllability makes semiconductors the foundation of modern electronics.

The most widely used semiconductor is silicon. Semiconductor physics explains how charge carriers move, how a PN junction is formed, and how devices such as diodes, transistors, LEDs, solar cells and integrated circuits operate.

2. Objectives of the Mini Project

- Understand the basic physical properties of semiconductors.
- Explain energy bands and the significance of the band gap.
- Differentiate intrinsic and extrinsic semiconductors.
- Understand doping and the formation of P-type and N-type materials.
- Explain the formation and working of a PN junction.
- Understand the basic working principle of a semiconductor diode.
- Connect semiconductor physics with real-life and modern technologies.
- Develop conceptual clarity for presentation and faculty Q&A.;

3. Concept / Theory

In an isolated atom, electrons occupy discrete energy levels. In a solid, interactions between a very large number of atoms produce closely spaced allowed energy ranges called energy bands. The two important bands in semiconductor physics are the valence band and the conduction band. The energy region between them is the forbidden energy gap or band gap, E_g .

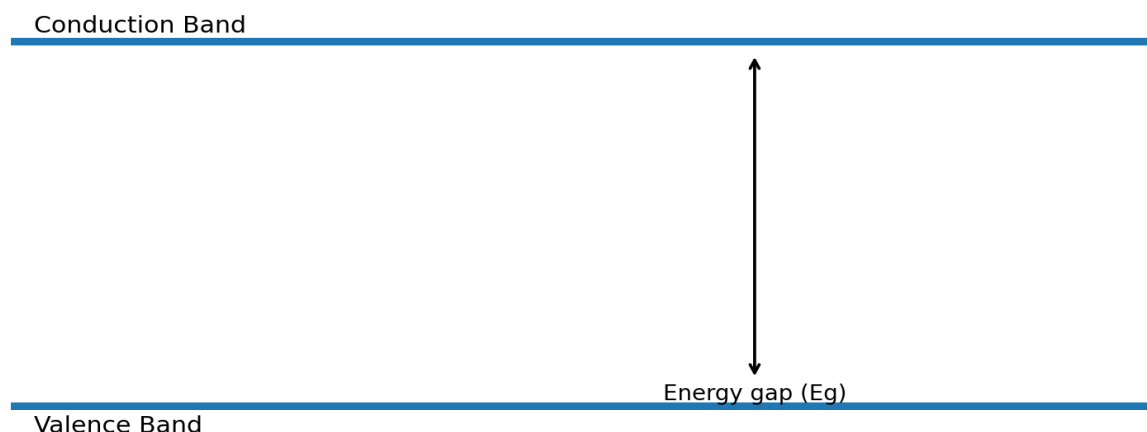


Figure 1. Simplified energy-band representation.

At ordinary temperatures, some electrons in a semiconductor can acquire enough energy to move from the valence band into the conduction band. This creates mobile electrons and leaves vacancies called holes in the valence band. Both electrons and holes can contribute to electrical conduction.

4. Why Are Semiconductors Special?

- Conductivity is higher than that of insulators but lower than that of metals.
- Their conductivity generally increases as temperature increases.
- Light can generate charge carriers in semiconductor materials.
- Controlled impurities can dramatically change conductivity.
- Electrical behaviour can be engineered to create rectifiers, switches, amplifiers, light emitters and sensors.

5. Intrinsic Semiconductor

A pure semiconductor without intentionally added impurities is called an intrinsic semiconductor. In pure silicon, atoms form covalent bonds. Thermal energy can break some bonds, producing an electron-hole pair. In an intrinsic semiconductor, the number of free electrons and holes is equal.

6. Extrinsic Semiconductor and Doping

A semiconductor whose electrical properties are modified by adding a controlled amount of impurity is called an extrinsic semiconductor. The process is called doping. Doping increases the number of useful charge carriers and allows engineers to control conductivity.

P-type semiconductor: A trivalent impurity such as boron is introduced into silicon. It creates holes, so holes are the majority carriers and electrons are minority carriers.

N-type semiconductor: A pentavalent impurity such as phosphorus is introduced into silicon. It contributes extra electrons, so electrons are the majority carriers and holes are minority carriers.

7. P-Type vs N-Type

Feature	P-Type	N-Type
Dopant example	Boron (trivalent)	Phosphorus (pentavalent)
Majority carriers	Holes	Electrons
Minority carriers	Electrons	Holes
Overall material	Electrically neutral	Electrically neutral

8. PN Junction

When P-type and N-type semiconductor regions are joined, a PN junction is formed. Near the junction, electrons from the N-side diffuse toward the P-side and holes from the P-side diffuse toward the N-side. Recombination leaves behind fixed charged ions, creating a region with very few mobile charge carriers called the depletion region.

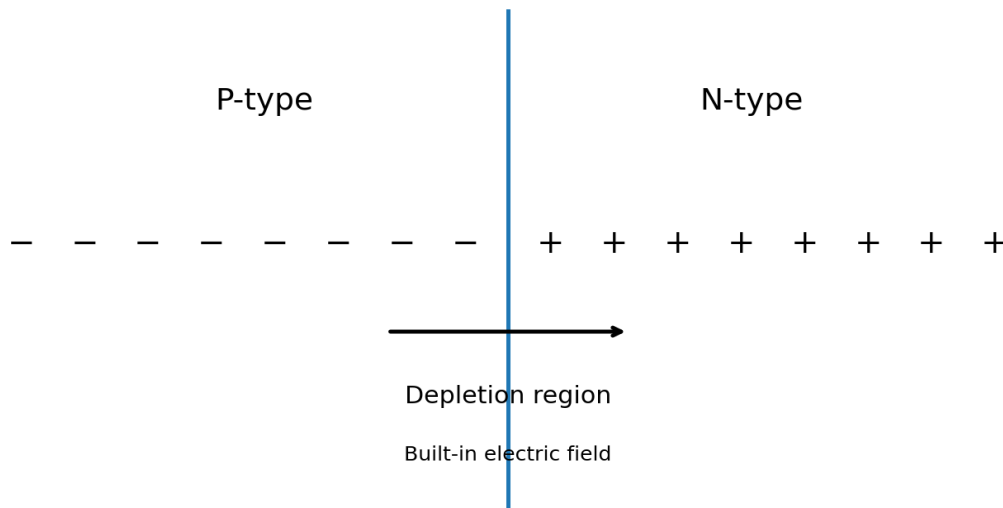


Figure 2. Simplified PN junction and depletion region.

9. Working Principle of a Semiconductor Diode

A diode is a semiconductor device based on a PN junction. Its main function is to allow current to pass readily in one direction while strongly restricting current in the opposite direction.

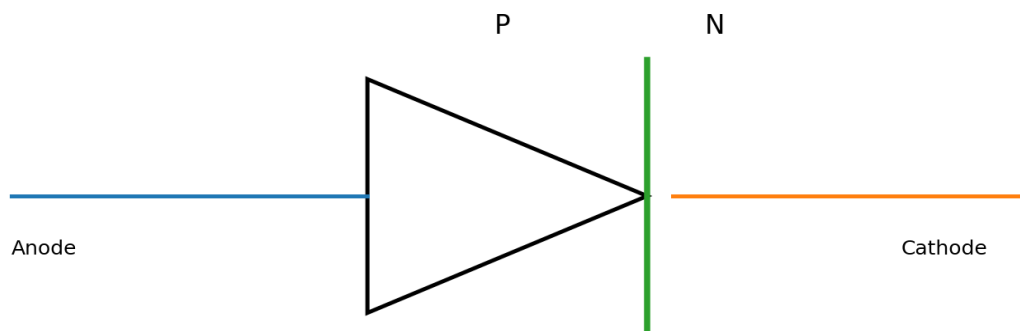


Figure 3. Simplified PN-junction diode symbol and terminals.

- Forward bias: P-side is connected to the positive terminal and N-side to the negative terminal. The depletion region becomes narrower and current can flow more readily after the junction barrier is sufficiently reduced.
- Reverse bias: P-side is connected to the negative terminal and N-side to the positive terminal. The depletion region widens and only a very small leakage current normally flows until breakdown conditions are reached.

10. Important Principles / Equations

- Drift and diffusion: Charge carriers can move due to electric fields (drift) and concentration differences (diffusion).
- Conductivity: Semiconductor conductivity depends on the concentration and mobility of charge carriers.
- Diode current relation: $I = I_0 [e^{(qV/\eta kT)} - 1]$ (idealized diode equation; symbols depend on convention).
- Energy relation: Photon energy $E = hf$, relevant to optoelectronic semiconductor devices.
- Band gap: E_g represents the energy separation between valence and conduction bands.

For a Class 10/introductory presentation, the most important relationship to remember is the physical idea: controlling charge carriers controls conductivity.

11. Working Flow of Semiconductor Technology

- Start with a semiconductor material such as silicon.
- Control its electrical properties through doping.
- Create P-type and N-type regions.
- Join regions to form PN junctions and semiconductor devices.
- Use controlled current, voltage, light or charge to perform a desired function.
- Integrate millions or billions of devices into modern electronic systems.

12. Real-Life Applications

- Diodes: Rectification, protection and signal control in electronic circuits.
- Transistors: Electronic switching and amplification; fundamental building blocks of digital logic.
- LEDs: Semiconductor devices that convert electrical energy into light.
- Solar cells: Semiconductor junctions convert light energy into electrical energy.
- Integrated circuits: Large numbers of semiconductor devices are fabricated on tiny chips.
- Sensors: Semiconductor materials respond to light, temperature, magnetic fields, gases and other physical quantities.
- Power electronics: Semiconductor devices control and convert electrical power in chargers, inverters and electric systems.

13. Advantages and Limitations

Advantages	Limitations / Challenges
Small size and high integration	Sensitive to excessive heat
Low power consumption in many applications	Manufacturing requires high precision
Fast switching and signal control	Performance depends on material quality
Conductivity can be engineered by doping	Very high voltages/currents require specialized devices

14. Recent / Modern Applications

Semiconductor technology is central to modern computing, smartphones, communication systems, artificial intelligence hardware, electric vehicles, renewable-energy systems and advanced sensing. Modern chips integrate enormous numbers of transistors at microscopic scales. Wide-bandgap materials such as silicon carbide (SiC) and gallium nitride (GaN) are increasingly important for efficient power conversion and high-frequency electronics. Semiconductor-based sensors and optoelectronic devices are also widely used in automation, healthcare instrumentation, imaging and communication.

15. Example: How a Solar Cell Uses a Semiconductor

A solar cell uses a semiconductor junction to convert light into electrical energy. When photons with sufficient energy are absorbed, they can create electron-hole pairs. The built-in electric field of the junction helps separate these charge carriers, producing a voltage and allowing useful current to flow through an external circuit.

16. Conceptual Q&A; / Viva Preparation

Q: What is a semiconductor?

A: A material whose electrical conductivity lies between that of a conductor and an insulator and can be controlled significantly.

Q: Why is silicon widely used?

A: It is abundant, has suitable semiconductor properties, forms a stable oxide useful in manufacturing, and can be processed reliably.

Q: What is doping?

A: The controlled addition of a small amount of impurity to modify a semiconductor's carrier concentration and conductivity.

Q: What is P-type?

A: A semiconductor doped with a trivalent impurity so holes become the majority carriers.

Q: What is N-type?

A: A semiconductor doped with a pentavalent impurity so electrons become the majority carriers.

Q: What is a PN junction?

A: The boundary formed when P-type and N-type semiconductor regions are joined.

Q: What is depletion region?

A: The region around a PN junction containing very few mobile charge carriers because of recombination and exposed fixed ions.

Q: Why does a diode conduct mainly in one direction?

A: Forward bias reduces the junction barrier and allows substantial current, while reverse bias widens the depletion region and strongly limits current under normal conditions.

Q: What is the difference between conductor and semiconductor?

A: Conductors have many freely available charge carriers and high conductivity; semiconductor conductivity is lower and can be strongly controlled by temperature, doping and other conditions.

Q: Where are semiconductors used?

A: In diodes, transistors, LEDs, solar cells, integrated circuits, sensors and power-electronic devices.

17. Conclusion

Semiconductors are the physical foundation of modern electronics. Their special electrical behaviour comes from their energy-band structure and the controlled movement of electrons and holes. Through doping, P-type and N-type materials can be engineered, and joining them creates PN junctions that enable devices such as diodes, transistors, LEDs and solar cells. The ability to precisely control semiconductor behaviour has transformed computing, communication, energy systems and everyday technology.

18. References

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